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Molecular Insights of Hyaluronic Acid-Molnupiravir Conjugate as a New Drug in Targeting SARS-CoV-2 Viral Proteins

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ABSTRACT: The antiviral drug Molnupiravir (MOL) and its conjugate with Hyaluronic Acid (HA-MOL) demonstrate the significance of targeting the main protease (Mpro) and papain-like protease (PLpro) of SARS-CoV-2. The pure drug Molnupiravir and the designed polymer-drug conjugate using Molnupiravir and hyaluronic acid were evaluated through molecular docking, pharmacokinetic analysis, molecular dynamics studies, and Density Functional Theory (DFT) studies, focusing on their ability to inhibit Mpro and PLpro proteins, which are essential for SARS-CoV-2 replication and transmission. Results indicate that the HA-MOL conjugate exhibits superior activity against these target proteins compared to the pure drug. ADMET analysis further showed that the HA conjugate has enhanced pharmacokinetic properties over the free form of the drug. Molecular dynamics simulation studies of Molnupiravir and HA-MOL revealed robust MD simulation trajectories with the SARS-CoV-2 target proteins. DFT analysis indicates that the polymer-conjugated drug significantly enhances electrophilicity in the gas phase and stabilizes the drug in aqueous environments, supporting its effectiveness in biological systems for drug delivery. This study concludes that the Hyaluronic Acid-Molnupiravir conjugate is a promising new therapeutic that may offer enhanced control over the SARS-CoV-2 virus.

KEYWORDS: Molnupiravir, Hyaluronic acid, Molecular docking, ADMET, Molecular Dynamic Simulation, drug conjugation, DFT analysis

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1. INTRODUCTION

Since 2019, the respiratory disease has spread and caused a global epidemic [1]. The main symptoms include fever, which damages the lungs, causes multi-organ failure, and increases mortality. The novel coronavirus disease (COVID-19) caused a pandemic, which alarmingly increased morbidity and mortality and severely affected the global population [2,3]. The World Health Organization (WHO) reported 776 million cases affected by COVID-19 [4]. The mutational changes that occurred in SARS-CoV-2 exhibited structural modifications leading to an

increase in the rate of spread, viral infection needs to improve diagnostic methodology, treatment strategies, and public health [5,6]. Recently, around 4000 genetic modifications were reported in the coronavirus by Chavda et al., [7]. The distribution of the virus is due to the multiple mutations [8].

Molnupiravir is considered as an effective antiviral drug to treat COVID-19, and it is metabolically transferred into active metabolites in the blood plasma of the host. This drug

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interacts with the molecules responsible for causing avian influenza by forming hemagglutinin-neuraminidase and RNA-dependent RNA polymerase (RdRp), main protease (Mpro), and spike protein of COVID-19 [9]. Molnupiravir can act as a prodrug to completely arrest the coronaviral replication process in the culture of human cell lines and experimental animal models [10]. In the treatment strategies, Molnupiravir drastically reduced the hospitalization risk and mortality during the pandemic. Several researchers have reported on the viral polymerase enzyme binding and broad-spectrum antiviral activity of Molnupiravir against human seasonal coronaviruses.

For the past decades, conjugation chemistry and binding affinity of hyaluronic acid and its drug delivery potential have been studied by several researchers [11-14]. The present study aims to develop a cell-targeted drug delivery (CTDD) system based on Hyaluronic Acid (HA), a natural polymer that appeared ubiquitously in the human system, making it a drug carrier to target and deliver the drugs for the treatment of diseases caused by viruses, especially SARS-CoV-2 or the disease includes tuberculosis and many types of cancers [15,16]. A cluster of differentiation 44 (CD44) is a glycoprotein observed on the cell surface involved in the expression of cancers like brain, lung, pancreatic, ovary, and breast cells. It exhibits a function on the cell's surface and participates in cell proliferation, interaction between cells, migration of cellular components, inflammatory

response, and related immune responses. It has a selective binding affinity with hyaluronic acid [17].

Figure 1 (SARS-CoV-2 and HA-MOL internalization) depicts the main biological stages required for the HA-MOL drug conjugate to deactivate and eradicate SARS-CoV-2 from the cell. The SARS-CoV-2 virus enters cells through the Angiotensin-Converting Enzyme 2 (ACE2) receptor-mediated endocytic pathway. This pathway consists of dynamic membrane-enclosed tubule-vesicular structures with various biological functions. These functions including early endosomes, late endosomes, recycling endosomes, and lysosomes.

- The HA-MOL medication enters the cell via the CD44 receptor.
- Hydrolytic enzymes produced from the lysosome split the HA-MOL ester covalent link, releasing the medication into the cell.
- Glycosylation is a highly controlled system in cells that affects protein structure, function, recruitment, and signaling properties.
- The autophagosome initiates the virus's catabolic process, delivering cytoplasmic material to lysosomes for degradation of viral protein aggregates and facilitating bioenergetic homeostasis. The degradation components can be excreted out of the cell via exocytosis or recycled to produce new cells.

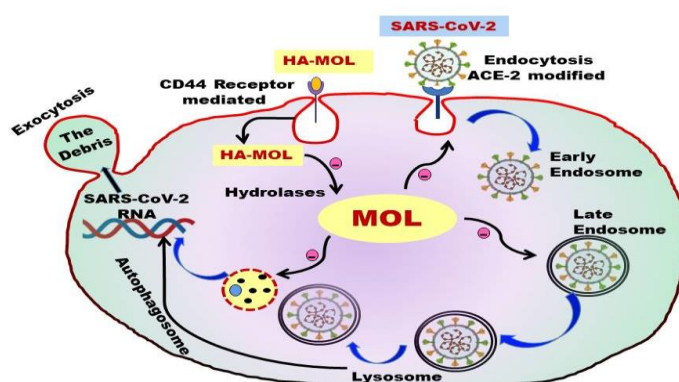


Figure 1: Key Biological Steps involved in cell targeted drug delivery (CTDD) pattern of HA-MOL conjugate

Density functional theory (DFT) is an important method based on quantum theory for the calculation of the electronic structures of atoms and molecules. Researchers study the

comprehensive electronic properties of drug molecules during drug design research to better understand the interaction between the drug and the receptor. Chemical accuracy is attained in this method, which is not reached by the

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molecular mechanics study, and it describes the properties of the drug and the inhibition rate on drug targets. In COVID -19 researches, the roles of molecular mechanics and DFT studies are more effective, and it is considered to be the sole technology to perform the calculations of the electronic structure engaged to find the interactions between the drug and its targets [18]. The molecular and electronic structure of the drug camptothecin, an anticancer drug, was studied by Subramanian et al. [19]. DFT can be described for the pharmacological analysis and the inhibition effects on the drug targets and the biomolecular systems clearly depicted by the DFT analysis [20].

The study carried out will include: (1) antiviral drugs, Molnupiravir and their HA-conjugated form of a drug targeted to the viral pathogen (2) the HA-drug conjugate designed by incorporating the specific functional group, e.g., -OH and other groups like an amino group, to the C-5' position of the glucuronic acid moiety to produce HA-MOL conjugate via an ester linkage if needed (Figure 2) to study through the *in-silico* analysis of HA-drug conjugate and its effectiveness on the main protease (Mpro) and papain-like protease (PLpro) available in SARS-CoV-2. Main protease (Mpro) effectively participates the viral infection and replication. It can be used as an appropriate target and involved in viral therapeutics [21]. Papain-like protease (PLpro) is the vital protein important

to form the replica transcriptase complex [22], and it plays a peculiar role in viral replication and making a crucial role in the pathogenetic pathway. This protein mainly accommodates the insertion of the host membranes and interacts with the non-structural viral proteins [23]. The key objectives of the current study are to find out a comparative study of Molnupiravir and its hyaluronic acid conjugate with its SARS-CoV-2 replication target proteins. Further, the efficacy of HA-MOL conjugate towards SARS-CoV-2 targeted delivery will be affirmed by various other *in-silico* screening tools, i.e., drug-likeness, ADMET, DFT, and molecular simulation studies reported.

2. MATERIALS AND METHODS

2.1. Ligands Preparation

The 2D structures of the compound Molnupiravir (PubChem compound ID 145996610) and its hyaluronic acid conjugate HA-Molnupiravir (HA-MOL) were drawn via the ACD/ChemSketch online tool, and files were saved in mol format (ACD 2021) (Figure 2). The 2D mol files with ligand structures used for the current research work were retrieved and submitted to the Novopro web server for conversion into three-dimensional PDB files. (<https://www.novoprolabs.com/tools/smiles2pdb>). The PDB-formatted ligand files were obtained and used for computational studies.

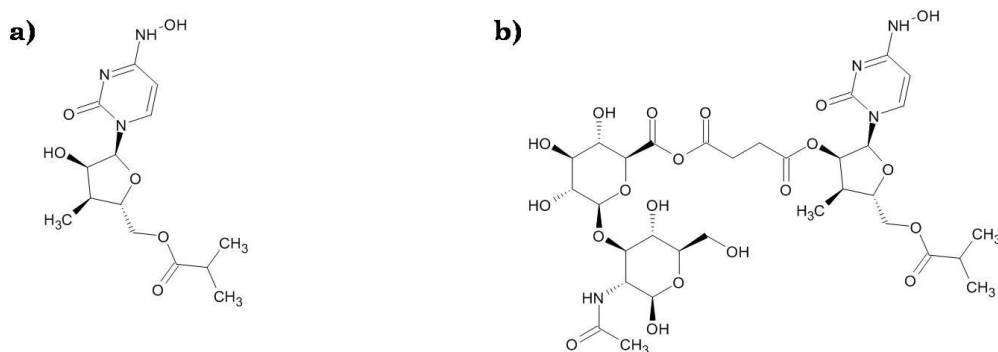


Figure 2: Structure of antiviral drugs a) Molnupiravir (MOL) (pubchem ID 145996610) b) HA-Molnupiravir (HA-MOL)

2.2. Preparation of Target Protein

The three-dimensional target protein structures are crucial for SARS-CoV-2 such as the COVID-19 main protease (PDB ID 6M03), the papain-like protease of SARS CoV-2 (PDB ID 6W9C), SARS-CoV-2 Omicron Mpro (PDB ID 7TVS), retrieved

from the protein data bank server. The 3D protein structures obtained in PDB format were optimized by removing heteroatoms and water molecules followed by energy minimization and protein protonation [24].

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2.3. Drug-likeness calculations and ADMET prediction

The Swiss ADME (www.swissadme.ch/) online server was used to examine the drug-likeness properties of the proposed drugs regarding COVID-19 [25]. pKCSMonline server (<https://biosig.lab.uq.edu.au/pkcsm/>) was used to screen the pharmacokinetic properties of proposed drugs regarding COVID-19. Drug effectiveness toward disease target proteins is assessed through its superior pharmacokinetic and drug-likeness properties, which influence the physicochemical properties and metabolism of drug molecules. The chemical structure of the drug was sufficient to validate its pharmacokinetic properties.

2.4. Bioactivity score prediction

The bioactivity score of a drug is important for identifying drug candidates with therapeutic potential against target diseases. The drugs, including Molnupiravir and its hyaluronic acid drug conjugate were analyzed through the molinspiration online server. Molinspiration servers are known for identifying receptors such as kinases, ion channels, GPCRs, proteases, nuclear receptors and enzyme inhibitors related to drug bioactivity measurements.

2.5. DFT study

Density Functional Theory (DFT) simulations through the Gaussian 09W software program were used to evaluate the molecular characteristics [26]. The Lee-Yang-Parr correlation functional and Becke's three-parameter exchange functional were integrated using the B3LYP density functional approach [27, 28]. The study was carried out by taking into account both the gas and aqueous solvent phase conditions at the B3LYP/6-31G(d, p) level of theory. The compounds under investigation varied; molecular and electronic properties were ascertained by utilizing the optimum outcomes of these DFT calculations.

2.6. Molecular docking study

AutoDock4 was used to perform molecular docking analysis of selected target ligands, and the AutoDock tool was used for molecular docking calculations. The grid box in X, Y, and Z dimensions was fixed in each target binding site for molecular docking analysis. Immobilization

of ligands to obtain 3D conformations via PyMOL and universal force fields were used. Then, it is saved in the PDBQT format. The top-scored protein-ligand complex conformations of the entire complex model were analyzed to evaluate the most binding and least binding affinity between ligand and target protein. The Discovery Studio visualizer 2.5 was used to obtain three-dimensional docking poses, and 2D interaction images of the protein-ligand complex and the results were analyzed [29].

2.7. MD simulation analysis

MD simulations were performed through Desmond software (version 3.6) for the following 4 complexes: 6M03-Molnupiravir, 6M03-HA-Molnupiravir, and 7TVS-Molnupiravir, 7TVS-HA-Molnupiravir. The best ligand conformers obtained from docking studies were used in MD experiments. Protein-ligand complexes were prepared for simulation via the protein preparation wizard tool with the Schrödinger Maestro preset parameters [30]. In the TIP3 model with PBC and OPLS force fields, sodium and chloride ions neutralize the body charge utilized for simulation studies [31]. Thus, the system completes the energy reduction in 150 ns, 2000 steps before production. MDS was performed via Berendsen temperature measurement and temperature measurement (NPT) assembly [32,33]. The amount of NaCl was adjusted to 0.20 M to simulate the body condition. MD simulation trajectories were recorded every 50 ps, and RMSD values were obtained to analyze the strength of the protein-ligand complex over time.

2.8. MM-GBSA analysis

Post embedding reduction was performed by importing placed complex poses via the MM-GBSA method [34]. Default configurations such as the OPLS3 force field was used [35]. The ΔG MM GBSA complex was designated as an effective ligand-receptor complex, and the receptor-ligand is the energy combination that determines the contribution of the free ligand and receptor, respectively. Schrödinger's thermal_mmgbsa.py script was used to find the MM-GBSA.

3. RESULTS AND DISCUSSION

3.1. Drug-likeness analysis

The drug-likeness results of the antiviral drug Molnupiravir and its hyaluronic acid conjugate

were validated, and the findings are summarized in Table 1. To assess the drug-likeness properties of these molecules and evaluate their quality and potential usefulness, Lipinski, Muegge, Egan, Ghose, and Veber rules were applied. Molnupiravir, both in its free and conjugated forms, showed no violations of these drug-likeness rules.

Physicochemical properties were calculated for each molecule to verify their suitability for drug formulation, with no violations observed

across the assessed parameters. However, further drug testing is required to confirm drug-likeness in practical applications. An additional analysis using the MRP1 method was performed to assess the topological polar surface area (TPSA), which indicates the molecule's transport capability. Compounds with higher TPSA values are more likely to reach target sites effectively, while those with lower values may not. In this study, compounds with TPSA values of 150 or lower and up to nine exchangeable bonds were identified as a potential drug.

Table 1: Drug likeliness screening of antiviral drug Molnupiravir and its Hyaluronic acid drug conjugate

Compound	HA	RB	HBA	HBD	MR	TPSA	XLOGP3	WLOGP	MLOGP	Lipinski violation	Ghose violation	Veber violation	Egan violation	Muegge violation
Molnupiravir MW=327.33	23	6	7	3	79.67	122.9	-0.24	-0.38	-0.12	0	0	0	0	0
HA-Molnupiravir MW=806.72	56	19	21	9	176.8	350.5	-3.85	-5.21	-3.45	3	4	2	1	6

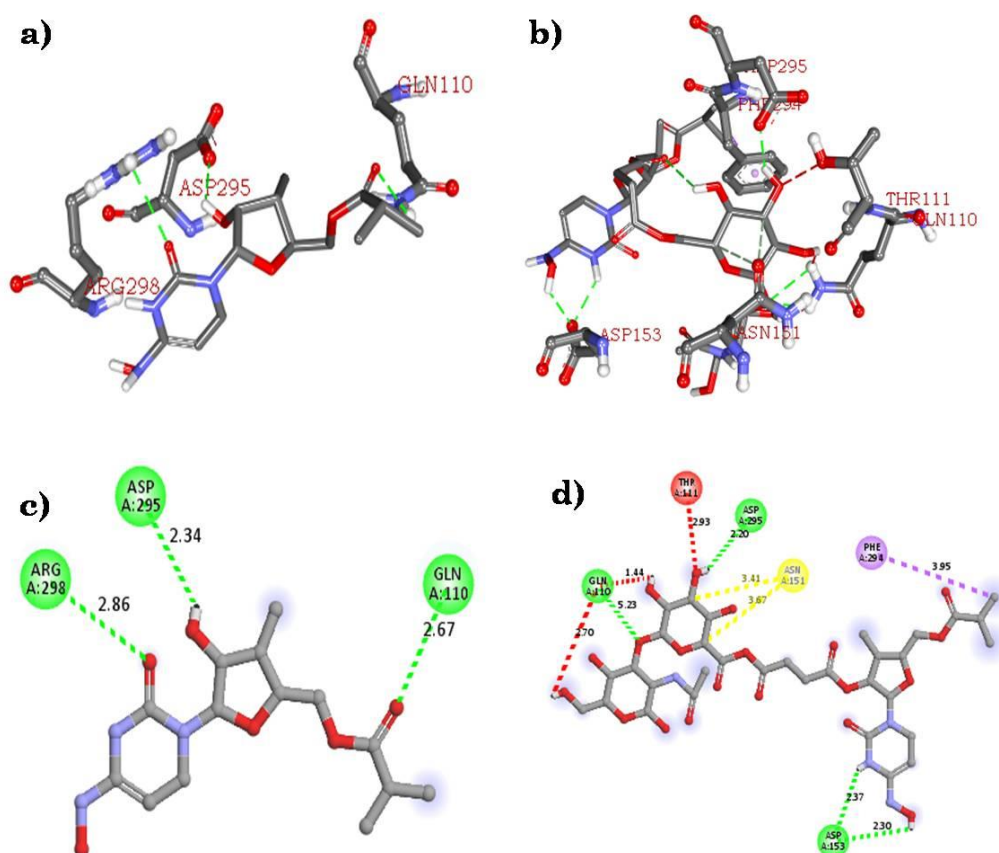
3.2. Molecular Docking

Molecular docking is an important method employed in the designing of novel drugs before chemical synthesis, and it helps to identify the conformational changes and observe the orientation of the drugs which in turn is proposed to bind the targeted molecular sites for drug development. The results of docking scores for the antiviral drugs Molnupiravir and its Hyaluronic acid conjugate against Mpro and PLpro of SARS-CoV-2 were examined by Autodock, as reported in Table 2 and Figures 3 to 5. The antiviral drug MOL and its conjugate HA-MOL displayed a docking score between -6.3 and -7.9 kcal/mol. The antiviral drug MOL and HA conjugated drug HA_MOL, showed strong binding affinity against all experimental target proteins. Table 2 shows the interactions that occurred between protein-ligand complexes and their docking scores. Docking scores of -6.4 and -6.9 kcal/mol have resulted in the respective drugs MOL and HA-MOL being targeted to the COVID-19 protein 6M03. The drug Molnupiravir docked with the 6M03 resulted in a strong hydrogen-bonded interaction with the residues Arg298, Asp295, Gln110 and there was no non-

bonded interaction. The HA-conjugated Molnupiravir drug bonded with the 6M03 resulting in only hydrogen-bonded interaction with the amino acid residues of Ala176, His175, Tyr154, Arg82, His73, Thr75 and there were no non-bonded interactions. The molecular binding between PLpro 6W93 with the experimental drugs including MOL and HA-MOL resulted the docking score of -6.3 and -6.7 kcal/mol. The binding of 6W93 and MOL showed hydrogen-bonded interaction with three residues Tyr264, Tyr273 and Thr301 and two non-bonded interactions with the residues Pro258, and Pro247. The protein 6W9C-HA molnupiravir resulted in only hydrogen-bonded interactions with six residues listed as Ala176, His175, Tyr154, Arg82, His73, and Thr75. There was no non-bonded interaction noted. The free drug targeted with the enzyme 7TVS showed only hydrogen bonded interactions with the residues of Asp153, Asn151, and Gln110. The binding between 7TVS-HA-MOL also resulted in only hydrogen-bonded interaction with the residues Leu287, Thr199, Asn238, Asp197, and Arg131. The drug Molnupiravir strongly arrests the multiplication of viral particles by insertion of errors into the SARS CoV-2 genome [36].

Table 2: Docking score and interaction plot for SARS-COV-2 drugs and its hyaluronic acid conjugate against different SARS-CoV-2 target proteins

Compounds	Docking Score (kcal/mol)	Hydrogen Bonded Interaction	Non Bonded Interaction
6M03-Molnupiravir	-6.4	Arg298, Asp295, Gln110	-
6M03-HA Molnupiravir	-6.9	Gln110, Thr111, Asp295, Phe298, Asp153	Asn151
6W9C -Molnupiravir	-6.3	Tyr264, Tyr273, Thr301	Pro258, Pro247
6W9C -HA Molnupiravir	-6.7	Ala176, His175, Tys154, Arg82, His73, Thr75	-
7TVS -Molnupiravir	-7.1	Asp153, Asn151, Gln110	-
7TVS -HA Molnupiravir	-7.6	Leu287, Thr199, Asn238, Asp197, Arg131	-

**Figure 3: Three dimensional docking pose and two dimensional interaction plot for (a & c) 6M03-Molnupiravir (b & d); 6M03-HA-Molnupiravir**

The PLpro of COVID-19 is an active peptidase enzyme consisting of a catalytic triad with amino acid residues like Asp287, Cys112, and His273 which are strongly engaged in the viral replication process and pathogenic potential [23]. In the PLpro, Cys112 functions as a nucleophile, Asp287 is joined with histidine, and the residue His273 acts as an acid-base, and the complete process carried out by cysteine protease also involves catalytic activity [37].

However, in the case of Remdesivir, it is recognized as an adenosine analogue to inhibit both Middle East Respiratory Syndrome (MERS) and Severe Acute Respiratory Syndrome (SARS), and COVID -19 [38,39]. Remdesivir is also controlling the replication of viral particles [40,41]. The *in-silico* analysis reported by Mohan et al., [42] highlighted that the hyaluronic acid conjugated Dexamethasone and Favipiravir with the viral enzymes Mpro and PLpro recorded

better docking scores with the comparison of the viral drug alone. Moreover, Atazanavir-Hyaluronic acid conjugate (HA-ATZ) also resulted in a better docking score with the target COVID-19 protein molecules ACE-2, Mpro, and PLpro [43]. In addition to its inherent medicinal qualities, hyaluronic acid is notable for its biocompatibility, bioabsorbability, and capacity to promote medication transport inside the body. High molecular weight hyaluronan, for example, has been demonstrated to enhance embryogenesis, and aid in tissue regeneration, wound healing, and epithelial integrity maintenance while suppressing cell differentiation. On the other hand, lesser molecular weight hyaluronic acid has immunosuppressive and anti-angiogenic

properties that impede differentiation by preventing ligand access to cell receptors and reducing cell-cell interactions [44,45,46]. Muthuraj et al., [47] reported that the binding efficiency of the drug lopinavir potentially targeted the catalytic dyad, His41 - Cys145 of the SARS-CoV-2 main protease.

The catalytic triad of Cys112, His273, and Asp287 in the PLpro enzyme of COVID-19 plays a crucial role in viral replication and pathogenesis [37]. PLpro exhibits cysteine protease catalytic activity, where Histidine 273 (His273) functions as an acid-base, Cysteine 112 serves as a nucleophile, and Asparagine 287 is linked to histidine to facilitate the incorporation and deprotonation process involving Cysteine 112 [48].

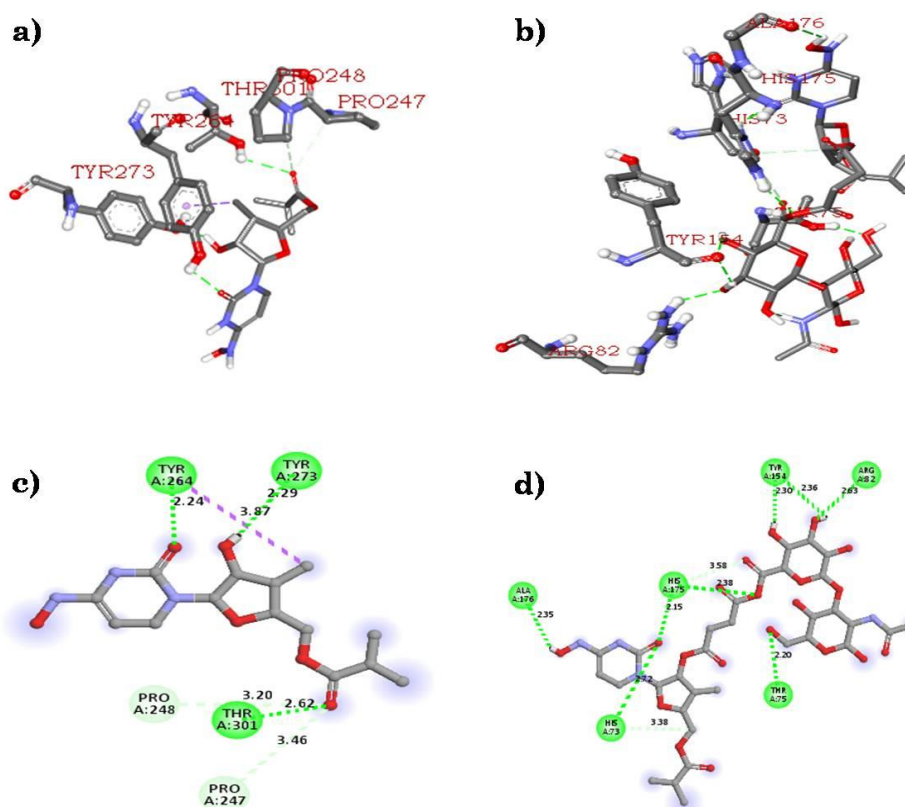


Figure 4: Three dimensional docking pose and two dimensional interaction plot for (a & c) 6W9C-Molnupiravir (b & d) 6W9C-HA-Molnupiravir

3.3. ADMET Analysis

The results of the ADMET analysis of the SARS-CoV-2 drug molnupiravir and its hyaluronic acid conjugate are reported in Table 3. The absorption of the drug molnupiravir and their its hyaluronic acid conjugate showed human intestinal absorption (HIA) values of 74.60 and 72.97, respectively. Drug absorption was

measured by various factors such as intestinal absorption, phosphoglycoprotein substrate inhibitors, cell membrane, and skin permeability. All pharmacokinetic properties of these compounds were analyzed using SWISS ADME software.

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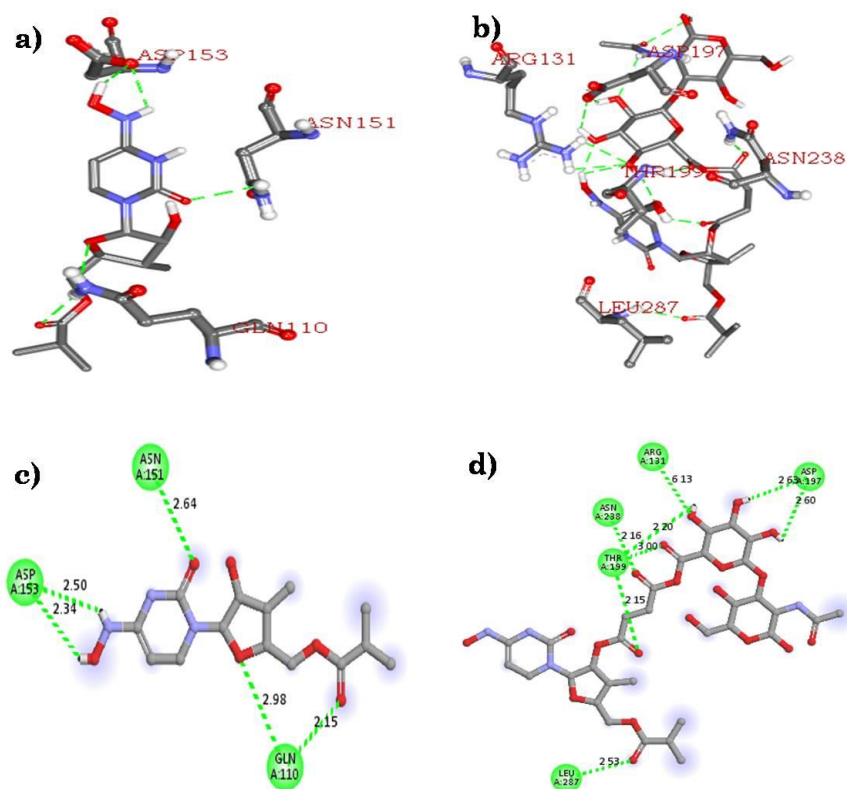


Figure 5: Three dimensional docking pose and two dimensional interaction plot for (a & c) 7TVS-Molnupiravir (b & d) 7TVS-HA-Molnupiravir

Table 3- ADMET Properties of Molnupiravir and its Hyaluronic acid- Molnupiravir conjugate

ADMET	Molnupiravir	HA-Molnupiravir
ABSORPTION		
Caco-2 cell permeability (10^6 cm/sec)	0.194	0.916
P-glycoprotein I Inhibitor	No	No
P-glycoprotein II Inhibitor	No	No
Skin Permeability log Kp	-2.747	-2.735
Water Solubility log/mol/L	-2.697	-2.556
HIA	74.60	72.77
DISTRIBUTION		
CNS Permeability	-3.235	-6.315
Blood-brain barrier penetration (log BB)	-0.964	-2.9
Fraction Unbound (Fu)	0.683	0.412
METABOLISM		
CYP1A2 Inhibition	No	No
CYP2C9 Inhibition	No	No
CYP2C19 Inhibition	No	No
CYP2D6 Inhibition	No	No
CYP3A4 Inhibition	No	No
EXCRETION		
Total Clearance (log ml/min/kg)	0.304	0.720
Renal OCT2 Substrate (log mg/kg/day)	No	No
TOXICITY		
AMES Toxicity	No	No

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Minnow Toxicity log mM	2.679	10.554
hERG I Toxicity	No	No
hERG II Toxicity	No	Yes
Oral Rat Acute Toxicity (LD50) (mol/kg)	2.021	2.533
Hepatotoxicity	Yes	No
Skin Sensitization	No	No

Hyaluronic acid conjugated with Molnupiravir shows the highest CaCo-cell permeability values of 0.916×10^6 cm/sec, but the pure drug molnupiravir possesses 0.194×10^6 cm/sec. Skin permeability values of -2.747 log Kp and -2.735 log Kp for the drugs MOL and HA-MOL respectively. The water solubility of -2.697 log/mol/L and -2.556 log/mol/L was observed with the drugs MOL and HA-MOL, respectively. The examination of central nervous system permeability, bioavailability, blood-brain barrier (BBB) penetration, and unbound fractions of drug molecules was cleared and validated through ADMET analysis. The analysis of ADMET properties is used to consider the extent and rate of absorption of drugs by the cells of the intestine and confirm the bioavailability of the drugs [49]. Caco-2 cell permeability is considered to be an essential property to demonstrate the gut-blood barrier [50]. The absorption properties of drugs can be analyzed through the values of HIA and Caco-2. The maximum percentage of metabolic reactions around ninety percent was estimated by the CYP proteins, like 1A2, 2C9, 2C19, 2D6, and 3A4 [51].

The drug molecule MOL and the conjugate HA-MOL exhibited no inhibition noted with the enzymes involved in metabolism listed as CYP1A2, CYP2C9, CYP2C19, CYP2D6, and CYP3A4. The total clearance of MOL and HA-MOL resulted from the clearance values of 0.304 log ml/min/kg and 0.720 log ml/min/kg, respectively. There is no AMES toxicity observed with experimental drugs.

Before clinical trials, it was noted that the ADMET properties of drug molecules play a major role in the identification of capable drug candidates for appropriate target proteins of any disease. Drugs scoring and possessing good ADMET properties have a higher success rate in clinical trials than drugs with poor ADMET properties. A plethora of recent research was reported on the use of conjugated drugs for diseases like tuberculosis [16,32,52,53] and SARS-CoV-2 [42,43,54] that resulted from their efficiency to control diseases. The ADMET profile of drug molecules envisages the key features of drug candidates and provides support to design appropriate drugs with safety concerns.

Table 4: Bioactivity score for antiviral drug molnupiravir and its Hyaluronic acid conjugate

Compounds	Parameters of Bioactivity score					
	GPCR ligand	Ion channel modulator	Kinase inhibitor	Nuclear receptor ligand	Protease inhibitor	Enzyme inhibitor
Molnupiravir	0.40	-0.06	0.11	-0.51	0.25	0.68
HA-Molnupiravir	-1.56	-2.96	-2.55	-2.82	-0.98	-1.68

3.4. Bioactivity Score

Bioactivity score is useful to identify drugs with specificity of ligand towards target protein possessing higher therapeutic index with less side effect, which could act as a novel drug candidate to target and control the proteins responsible for the disease. The bioactivity score of an antiviral drug molnupiravir and its hyaluronic acid conjugate was evaluated for its usefulness in ascertaining binding efficiency towards various target proteins such as Mpro

and PLpro. Table 4 shows the bioactivity scores for the proposed antiviral drug MOL and its conjugate HA-MOL. The bioactivity scores of the selected drugs can be categorized as biologically active (bioactivity score > 0.00), moderately active (bioactivity score between -5.0 and 0.00), and inert (bioactivity score > -5.0) [55]. As shown in Table 4, the bioactivity scores of an antiviral drug such as Molnupiravir and its hyaluronic acid conjugate possess positive values against some ions and channels predicted to be moderately active in all screened protein

targets, ions, and channels. The results of the present study concord with previous research findings [32,53,56]. G protein-coupled receptors (GPCR) hinder the surface of tumor cells and block their receptor activity. This will disturb the delivery of oxygen and nutrients to cells and interfere with the cell wall's ability to absorb and penetrate. According to Campillo et al., [57], the extract contains phytochemicals that alter cell roles, interact with tumor cells to lessen their cancer progression capacity, and finally lead to programmed cell death. An ion channel modulator obstructs and opens ions on the surface of the cell wall to prevent the growth of cancer cells. They can hamper the process of communication between and within cells, which can block the protein synthesis and stop the growth of tumor cells.

3.5. Molecular orbitals and electronic properties

According to frontier molecular orbital theory, chemical reactions typically occur at sites with high electron density and low energy levels. This

indicates that a molecule's reactivity and kinetic stability are significantly influenced by the distribution of intramolecular charge density [58]. The frontier molecular orbitals, with a focus on the Highest Occupied Molecular Orbital (HOMO) and the Lowest Unoccupied Molecular Orbital (LUMO), play crucial roles in determining a molecule's stability and reactivity [59]. While LUMOs have a strong electron affinity and acts as electron acceptors, HOMOs are characterized by their electron-donating properties and relatively weak hold on electrons [60]. The energy gap between the HOMO and LUMO, which represent the molecule's excitation energy, serves as a crucial indicator of chemical stability [61]. The electron cloud distribution for the molecules under study is illustrated in Figures 6a and 6b, demonstrating that electron density is dispersed across multiple atoms in both the HOMO and LUMO states. Understanding this distribution is essential for grasping molecular interactions during chemical reactions.

a.)

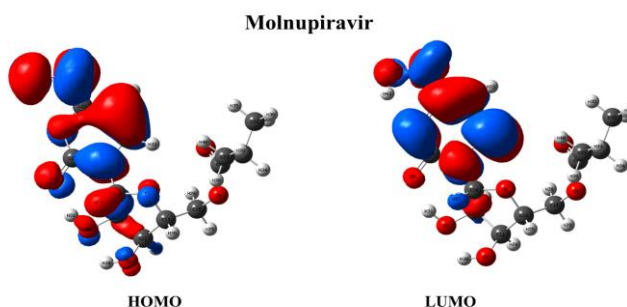


Figure 6a Electron cloud distribution diagram for Molnupiravir molecule

b.)

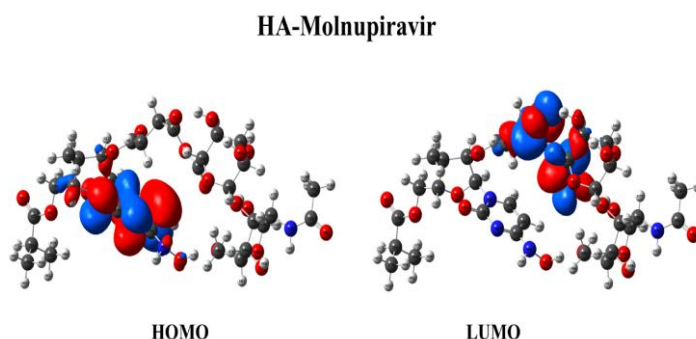


Figure 6b: Electron cloud distribution diagram for HA-Molnupiravir molecule

In general, a larger energy gap indicates greater stability, while a smaller gap suggests

higher reactivity. Therefore, the molecular structure of the title molecule appears relatively

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stable [62,63]. Table 5 show the molecular orbitals and electronic properties values for the optimized geometries of the Molnupiravir and HA-Molnupiravir molecules in the gas and water phase. According to Koopmans' theorem, the energy of the highest occupied molecular orbital (HOMO) in basic molecular orbital theory is equal to the negative value of a molecule's ionization potential (IP). This allows the ionization potential to be approximated using the HOMO energy (E_{HOMO}). The relationship is represented mathematically as follows [64].

$$IP = -E_{HOMO}$$

The electron affinity (EA) of a molecule can be calculated using the energy of the lowest unoccupied molecular orbital (LUMO). The following is how the electron affinity and LUMO energy (E_{LUMO}) are related:

$$EA = -E_{LUMO}$$

Additionally connected to electronegativity (χ), chemical hardness (η), and chemical softness (S) are the HOMO and LUMO energies [65].

$$\chi = \frac{(IP + EA)}{2}$$

$$\eta = \frac{(IP - EA)}{2}$$

$$S = \frac{1}{(IP - EA)}$$

According to Padmanabhan et al., [66], a species' chemical potential (μ) can be represented by its electron affinity (EA) and ionization potential (IP).

$$\mu = \frac{-(IP + EA)}{2}$$

The electrophilicity index (ω) measures whether a molecule accepts electrons and participates in chemical reactions as a global parameter of chemical reactivity [67]. It is defined as follows:

$$\omega = \frac{\mu^2}{2\eta}$$

The title molecules computed Molecular orbitals and electronic properties values are listed in Table 5.

Table 5: Molecular orbitals and electronic properties values for the optimized geometries of the Molnupiravir and HA-Molnupiravir molecules in the gas and water phase

Parameters	Molnupiravir		HA-Molnupiravir	
	Gas	Water	Gas	Water
HOMO (eV)	-6.207	-6.238	-6.178	-6.358
LUMO (eV)	-1.183	-1.164	-1.955	-1.421
HOMO-LUMO Gap (eV)	5.024	5.074	4.223	4.937
Electronegativity (eV)	3.695	3.701	4.067	3.890
Chemical hardness (eV)	2.512	2.537	2.112	2.469
Chemical softness (eV) ⁻¹	0.398	0.394	0.474	0.405
Chemical potential (eV)	-3.695	-3.701	-4.067	-3.890
Electrophilicity index	2.718	2.700	3.916	3.064

The electronic properties of Molnupiravir and its hyaluronic acid conjugate (HA-Molnupiravir) were analyzed in both the gas phase and aqueous solvent phase. For Molnupiravir in the gas phase, the HOMO energy was -6.207 eV, and LUMO was -1.183 eV, yielding an energy gap of 5.024 eV. The electronegativity (χ) was 3.695 eV, with a chemical hardness (η) of 2.512 eV, indicating moderate stability and resistance to electron density changes [68]. The chemical softness (S) was 0.398, the chemical potential (μ) was -3.695 eV, and the electrophilicity index was 2.717, suggesting moderate reactivity. In water, these values shifted slightly, with an HOMO at -6.238 eV and LUMO at -1.164 eV,

resulting in a minor increase in the energy gap to 5.074 eV. The chemical potential (μ) remained consistent at -3.701 eV, indicating stable electron-attracting ability.

Conversely, HA-Molnupiravir exhibited significantly altered properties; in the gas phase, its HOMO was -6.178 eV, and LUMO was -1.955 eV, leading to a reduced energy gap of 4.223 eV and an increased electronegativity of 4.066 eV, indicating enhanced reactivity. The chemical hardness decreased to 2.111 eV, while the softness increased to 0.474, reflecting greater flexibility. The chemical potential (μ) for HA-Molnupiravir in the gas phase was -4.0665 eV,

and the electrophilicity index rose to 3.916. In water, HA-Molnupiravir's HOMO and LUMO shifted to -6.358 eV and -1.421 eV, respectively (Figures 6a & 6b), with an energy gap of 4.937 eV, a slightly lower electronegativity of 3.889 eV, and a chemical potential (μ) of -3.889 eV. The chemical hardness increased to 2.468 eV, and the softness decreased to 0.405, indicating reduced reactivity compared to its gas phase properties. The HA conjugation significantly enhances the electrophilicity of Molnupiravir in the gas phase, and the aqueous environment stabilizes the conjugate, resulting in lower reactivity [69]. This suggests that HA-Molnupiravir may be more effective in biological systems for drug delivery while maintaining stability in physiological conditions [70,71].

3.6. Molecular Dynamic Simulation

Molecular dynamic simulation analysis was performed for 150ns for 6M03-Molnupiravir, 6M03-HA-Molnupiravir, 7TVS-Molnupiravir, and 7TVS-HA-Molnupiravir and the results were presented in Figures 7 to 10. The structural conformations identified through the analysis of RMSD values recorded with the protein-ligand

binding simulations gave rise to the protein dynamics. The root mean square deviation (RMSD) plots of the ligand structures have also been given to explain the stability of the protein binding pocket as plotted in Figure 7. The RMSD related to C-alpha atoms was calculated. The RMSD values supported monitoring the structural insights and simulation of the molecules and changes of 1-3 Å are acceptable for small globular proteins. The RMSD values of the present study stabilize around a fixed value. In the ligand RMSD analysis, it explained the stability of the ligand concerning the protein and its binding pocket. The protein 6M03 and drug Molnupiravir interaction showed deviation up to 40 ns simulation time and exhibited a maximum Root Mean Square Deviation value for protein at 0.8 Å and 3.2 Å. For the protein-ligand interaction for 6M03-HA-Molnupiravir, deviated up to 80ns simulation time and showed a maximum value of RMSD of protein at 1 Å and 2 Å. The 7TVS-Molnupiravir interaction attained maximum deviation up to 40ns simulation time and resulted in a maximum RMSD value of protein at 0.3 Å and 2.7 Å. Finally, the combination of the protein-ligand 7TVS-HA-Molnupiravir gave a deviation in simulation time of up to 80ns and found the maximum RMSD value of protein at 0.5 Å and 4.0 Å.

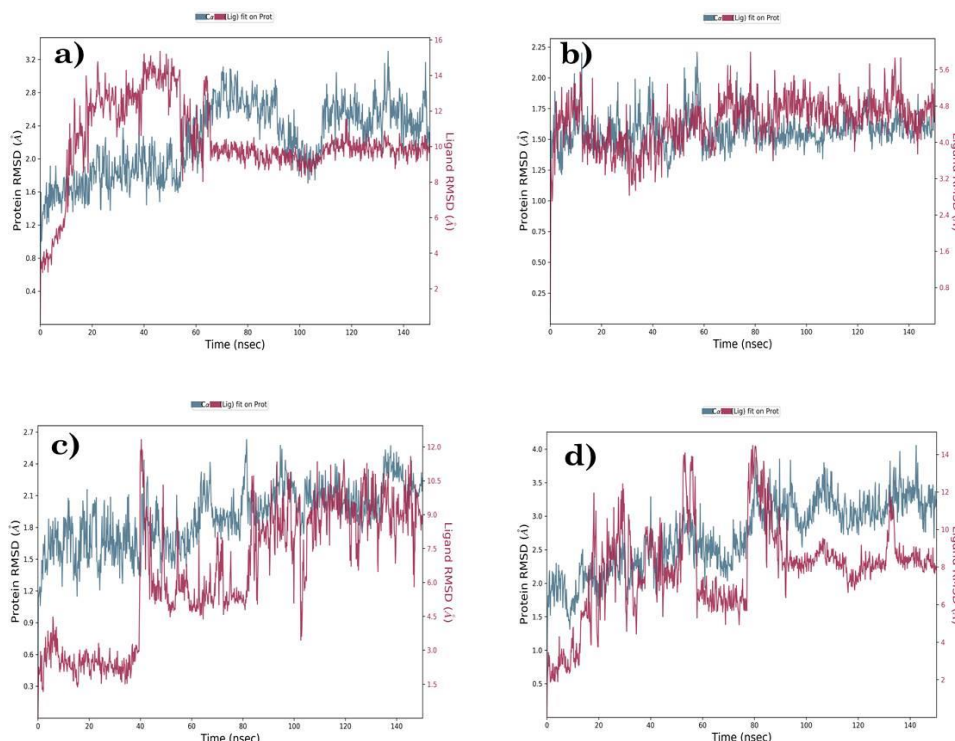


Figure 7: RMSD of the C-alpha atoms of the target protein and the ligand over time. The left y-axis shows the variation in the target protein RMSD, and the right y-axis shows the variation in the ligand RMSD over time a) 6M03-Molnupiravir b) 6M03-HA-Molnupiravir c) 7TVS-Molnupiravir d) 7TVS-HA-Molnupiravir

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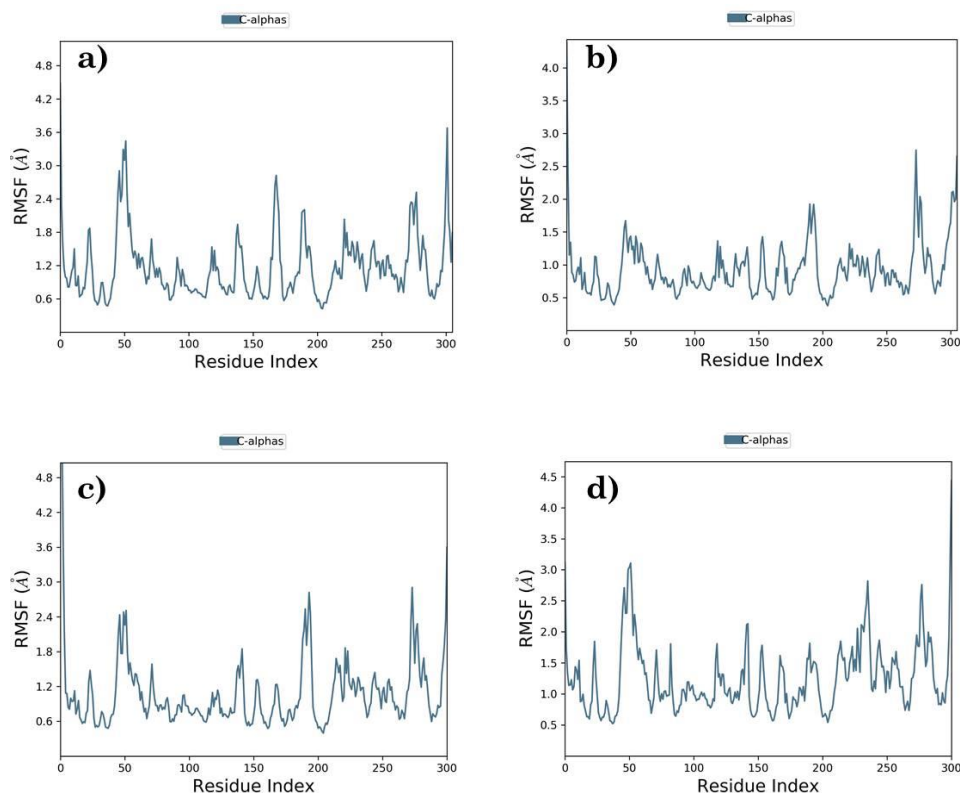


Figure 8: Residue-wise RMSF of ligand concerning receptor protein during triplicate simulations
a) 6M03-Molnupiravir b) 6M03-HA-Molnupiravir c) 7TVS-Molnupiravir d) 7TVS-HA-Molnupiravir

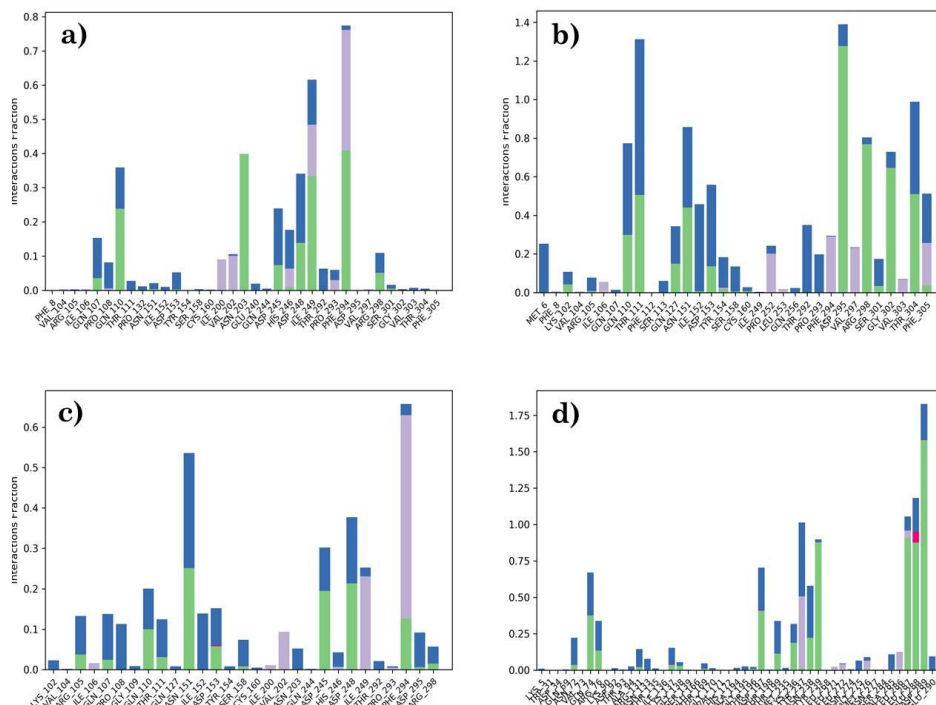


Figure 9: Protein-Ligand Interactions a) 6M03-Molnupiravir b) 6M03-HA-Molnupiravir c) 7TVS-Molnupiravir d) 7TVS-HA-Molnupiravir. The interactions are categorized as hydrogen bonds (Green), hydrophobic (Purple), ionic (Pink) and water bridges (Blue). The stacked bars are normalized over the course of the trajectory: for example, a value of 0.8 suggests that the specific interaction is maintained for 80% of the simulation.

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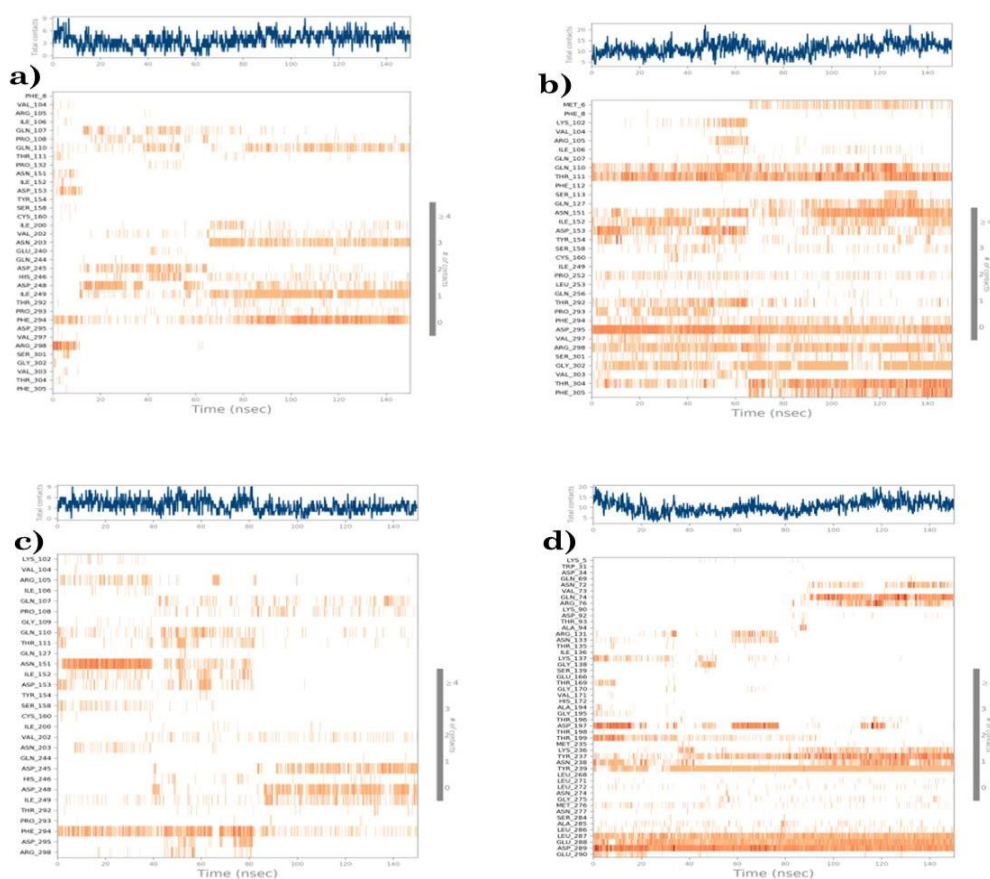


Figure 10: Time line representation of the interactions and contacts (H-bonds, Hydrophobic, Ionic, and Water bridges) the top panel of each image shows the total number of specific contacts the protein makes with the ligand throughout the trajectory. The bottom panel of each image shows which residues interact with the ligand in each trajectory frame. a) 6M03-Molnupiravir b) 6M03-HA-Molnupiravir c) 7TVS-Molnupiravir d) 7TVS-HA-Molnupiravir

The residue-wise Root Mean Square Fluctuation (RMSF) of the ligand concerned with the receptor protein is given in Figure 8. In the plot, the peak value indicated the areas of a protein fluctuated maximally during the simulation. The tail regions consisting of N and C terminals fluctuated more than the other parts of the protein observed in this study. The RMSF of the protein 6M03 with ligand Molnupiravir fluctuated between the range of 0.6 Å and 4.0 Å, for the protein 6M03 with ligand HA conjugated Molnupiravir fluctuated in 0.5 Å to 3.0 Å, for the protein interaction of 7TVS and Molnupiravir resulted 0.6 Å to 3.3 Å and for the protein 7TVS and HA conjugated Molnupiravir noted with 0.5 Å to 3.25 Å.

Alpha helices and beta strands are examples of protein secondary structural elements (SSE) that are monitored during the simulation. The distribution across the protein structure by residue index is shown by the SSE results. This information tracks the SSE assignment for every

residue over time and provides a summary of the SSE composition for every trajectory frame during the simulation. Based on the interactions between the proteins, alpha helices make up 14.72% to 18.30% and beta strands, 23.46% to 24.73%, for a total of 39.44% to 42.80%.

Throughout the simulation, the interactions between the protein and the ligand were observed and classified as hydrogen bonds, hydrophobic interactions, ionic bonds, and water bridges. An interaction study was done to find out what kind of bonding residues and ligands have. The simulated interaction diagram shown in Figure 9 provides additional precise subtypes of interaction found in this investigation. The normalized stacked bar charts over the trajectory demonstrate that values greater than 1.0 suggest that a protein residue may form several interactions of the same subtype with the ligand. The stable hydrogen bond with the residues Glu107, Glu110, Asn203, Asp245, Asp248, Ile249, Phe294, and Arg298,

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the hydrophobic interaction with the residues Ile200, Val202, His246, Ile149, Pro293, and Pro293, the water bridges between the residues Glu107, Pro108, Glu110, Thr111, Asp153, Glu240, Asp245, His246, Asp248, Ile249, Thr292, Pro293, Arg298, and Ser301 were observed in the protein 6M03 interacted with the ligand Molnupiravir (Figure 9a). The protein 6M03 interacted with the ligand HA-conjugated Molnupiravir, resulting in hydrogen bonding between the residues Thr304, Gly302, Arg298, Asp295, Asp153, Thr111, and Glu110, hydrophobic interaction with the residues Phe305, Val297, Phe294, Pro293, and Pro252, the water bridge interaction between the residues Met6, Gln110, Thr111, Gln127, Asn151, Ile152, Asp153, Tyr154, Ser158, Gln256, Thr292, Ser301, Thr304, and Phe305 (Figure 9b). Hydrogen bonding with the residues Arg105, Glu110, Asn151, Asp153, Asp245, Asp268, and Phe294, hydrophobic interaction observed with the residues of Val202, Ile249 and Phe294, water bridges noted between the residues Lys102, Asp105, Gln107, Pro108, Gln110, Thr111, Asn151, Asp1543, Ile152, Ser158, Asn203, Gln244, His2436, Asp248, Thr292, Asp295, and Arg298 were recorded with the interaction of the protein 7TVS with the ligand Molnupiravir (Figure 9c). Maximum hydrogen bonding resulted with the residues Asn72, Gln74, Asp197, Thr199, Tyr136, Asn238, Tyr239, Leu287, Glu288, and Asp289, hydrophobic interaction observed between the residues Tyr237, Met276, and Leu286, ionic interaction noted with the residue Glu288, water bridges were recorded with the residues Asn72, Glu74, Arg76, Arg131, Asn133, Lys137, Asp197, Thr199, Lys236, Tyr237, Asn238, Gly275, Ala285, Leu286, Glu 288, Asp289, and Glu290 were resulted in the interaction of the protein ligand 7TVS-HA-Molnupiravir (See Figure 9d). Further, the protein ligand contact graph was given in the Figure 10 and it shows the interaction of the particular residues at the specific time over the course of the simulation.

Recent studies showed that the Molecular dynamic simulation (MDS) experiments with drug molecules Masoprocol, Dobutamine, and Benserazide with the enzyme PLpro exhibited stability through RMSD and RMSF values of the experimental compounds [72]. The binding ability of Mpro with the drugs Ondansetron and Rolapitant and PLpro with the drug candidates Levomefolic acid and Labetalol showed good MD simulation during the studies [73]. In the

studies, it was inferred that the maximum inhibition rate is observed with the protein PLpro in the groove BL2 and loop regions constituting the amino acid residues Gln269 and Tyr268, and the conformational studies performed using MDS analysis [74].

3.7. MMGBSA Analysis

Results for MM GBSA analysis of an antiviral drug and its Hyaluronic conjugate, 6M03-Molnupiravir, 6M03-HA-Molnupiravir, 7TVS-Molnupiravir, 7TVS-HA-Molnupiravir depicted in Table 6.

Table 6: MM GBSA scores for MD simulated complexes

Complex	MMGBSA Score (-kcal/mol)
6M03-Molnupiravir	-94.28
6M03-HA-Molnupiravir	-103.71
7TVS-Molnupiravir	-79.53
7TVS-HA-Molnupiravir	-94.42

Complex 6M03-HA-Molnupiravir showed a maximal MMGBSA value-103.71kcal/mol than the free drug complex 6M03-Molnupiravir MMGBSA score -94.28kcal/mol. Likewise, hyaluronic conjugate 7TVS-HA-Molnupiravir showed a maximal MMGBSA value-97.42kcal/mol over free drug 7TVS-Molnupiravir value -79.53kcal/mol with target protein Omicron Mpro. MMGBSA analysis results confirm a greater binding affinity of the hyaluronic acid conjugate form of commercial SARS-CoV-2 drug against SARS-CoV-2 target proteins. Current study findings were found reliable with the previous study results [53].

CONCLUSION

Compared to free Molnupiravir (MOL), this molecular docking study aimed to assess the safety and effectiveness of a Hyaluronic Acid-Molnupiravir conjugate (HA-MOL). The study focused on SARS-CoV-2 viral protein targets and found that HA-MOL interacted more effectively with these targets than free drug. Notably, HA-MOL reduced host cell toxicity by enhancing drug transport to CD44 receptors in the lower respiratory tract. Molecular dynamics (MD) simulations further supported that HA-MOL-protein complexes were more stable than MOL-protein complexes. These results suggest that

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HA-MOL improves MOL's safety profile by reducing systemic toxicity and enhancing targeting. Additionally, electronic property analysis revealed significant changes in reactivity and stability for HA-MOL compared to Molnupiravir. In the gas phase, HA-MOL exhibited a reduced energy gap of 4.223 eV, reflecting enhanced reactivity, while in water, its energy gap was 4.937 eV, indicating stability in aqueous environments. HA-MOL also demonstrated higher electronegativity and electrophilicity in both phases compared to the free drug, suggesting that hyaluronic acid conjugation improves the drug's interaction potential and adaptability to physiological conditions. These electronic property changes align with improved drug delivery and targeting, making HA-MOL more effective in biological systems.

To further optimize HA-MOL's antiviral efficacy against key SARS-CoV-2 protein targets, such as Mpro and PLpro, ongoing *in-vitro* and *in-vivo* studies are exploring different levels of drug substitution in HA. This approach leverages HA-MOL's unique properties and its effective targeting of lower respiratory tract receptors, positioning it as a promising treatment option for COVID-19. In conclusion, this study highlights the importance of Hyaluronic Acid (HA) as a drug carrier to increase the bioavailability and efficacy of antiviral drugs. The proposed HA-MOL conjugate represents a novel therapeutic for combating COVID-19 and other viral infections through targeted drug delivery. Further studies are necessary to confirm these findings.

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Data availability

Data will be made available on request.

Declaration

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Conflict of interest

The authors declare no competing interests.

REFERENCES

- [1] Wu F, Zhao S, Yu B, Chen YM, Wang W, Song ZG, 2020. A new coronavirus

associated with human respiratory disease in China. *Nature*, 579, 265–9. <https://doi.org/10.1038/s41586-020-2008-3>

- [2] Bezbaruah, R., Chavda, V.P., Nongrang, L., Alom, S., Deka, K., Kalita, T., Ali, F., Bhattacharjee, B., Vora, L., 2022. Nanoparticle-Based Delivery Systems for Vaccines. *Vaccines* 10(11) 1946 <https://doi.org/10.3390/vaccines10111946>
- [3] Chavda, V.P., Balar, P., Vaghela, D., Solanki, H.K., Vaishnav, A., Hala, V., & Vora, L. 2023. Omicron Variant of SARS-CoV-2: An Indian Perspective of Vaccination and Management. *Vaccines* 11(1) 160. <https://doi.org/10.3390/vaccines11010160>
- [4] WHO (2024) <https://data.who.int/dashboards/covid19/cases?n=0> (accessed on 21st September 2024).
- [5] Bian, L., Gao, F., Zhang, J., He, Q., Mao, Q., Xu, M., Liang, Z., 2021. Effects of SARS-CoV-2 Variants on Vaccine Efficacy and Response Strategies. *Expert Rev. Vaccines* 20(4) 365–373. <https://doi.org/10.1080/14760584.2021.1903879>
- [6] Maurya, R., Mishra, P., Swaminathan, A., Ravi, V., Saifi, S., Kanakan, A., Mehta, P., Devi, P., Praveen, S., Budhiraja, S., 2022. SARS-CoV-2 Mutations and COVID-19 Clinical Outcome: Mutation Global Frequency Dynamics and Structural Modulation Hold the Key. *Front. Cell. Infect. Microbiol.* 12, 868414.
- [7] Chavda, V.P., Vuppu, S., Mishra, T., Kamaraj, S., Patel, A.B., Sharma, N., & Chen, Z.-S. 2022. Recent Review of COVID-19 Management: Diagnosis, Treatment and Vaccination. *Pharmacol. Rep.* 74, 1120–1148. <https://doi.org/10.1007/s43440-022-00425-5>
- [8] Celik, I., Tallei, TE., 2022. A computational comparative analysis of the binding mechanism of molnupiravir's active metabolite to RNA-dependent RNA polymerase of wild-type and Delta subvariant AY.4 of SARS-CoV-2. *J Cell Biochem.* Apr;123(4):807-818. <https://doi.org/10.1002/jcb.30226>
- [9] Istifli, E S., Okumus, N., Sarikurkcü, C., Kuhn, E R., Netz, P.A., Tepe, AS., 2024.

Thirumalaisamy et al.,

- Comparative docking and molecular dynamics studies of Molnupiravir (EIDD-2801): implications for novel mechanisms of action on influenza and SARS-CoV-2 protein targets. *J Biomol Struct Dyn.* 42(15) 8202-8214. <https://doi.org/10.1080/07391102.2023.2267696>
- [10] Wahl, A., Gralinski, L.E., Johnson, C.E. 2021. SARS-CoV-2 infection is effectively treated and prevented by EIDD-2801. *Nature*, 591, 451–457 <https://doi.org/10.1038/s41586-021-03312-w>.
- [11] Khan Riaz, Konowicz Paul A., Flaibani Antonella,, Gombac Valentina., 1997. Heavy metal salts of succinic acid hemiesters with hyaluronic acid or hyaluronic acid esters, a process for their preparation and relative pharmaceutical compositions, (1997) PAT: US6017901 <https://patentimages.storage.googleapis.com/87/82/80/98dd52d25f42ac/US6017901.pdf>
- [12] Sorbi. C., Bergamin, M., Bosi, S., Dinon, F., Aroulmoji, V., Khan, R., Murano, E., Norbedo, S., 2009. Synthesis of 6-O-methotrexyl hyaluronan as a drug delivery system. *Carbohydrate Research*, 344, 91-97. <https://doi.org/10.1016/j.carres.2008.09.021>
- [13] Khan, R., Mahendhiran, B., Aroulmoji V., 2013. Chemistry of hyaluronic acid and its significance in drug delivery strategies: A review. *International Journal of Pharmaceutical Sciences and Research*, 4, 3699-3710. [http://dx.doi.org/10.13040/IJPSR.0975-8232.4\(10\).3699-10](http://dx.doi.org/10.13040/IJPSR.0975-8232.4(10).3699-10)
- [14] Gnanendra, S., Rajesh, S.V., Desika, P., Syed, M., Sathiyapriya, R., Murano, E., Aroulmoji, V., 2017. Molecular insights of Hyaluronic acid as a potential source of drug conjugate in the target mediated treatment of cancer, *Natural Product Communications* 12(6) 813-819 (2017). <https://doi.org/10.1177/1934578X1801300501>
- [15] Norbedo, S., Dinon, F., Bergamin, M., Bosi, S., Aroulmoji, V., Khan, R., Murano E., 2009. Synthesis of 6-amino-6-deoxyhyaluronan as an intermediate for conjugation with carboxylate-containing compounds: application to hyaluronan - camptothecin conjugates. *Carbohydrate Research* 344: 1. 98-104. <https://doi.org/10.1016/j.carres.2008.09.027>
- [16] Thirumalaisamy Rathinavel, Palanisamy, M., Palanisamy, S., Subramanian, A., Thangaswamy Selvakumar ., 2020. Phytochemical 6-Gingerol – a promising Drug of choice for COVID-19, *Int. J. Adv. Sci. Eng.* 6 (2020) 1482–1489, <https://doi.org/10.29294/IJASE.6.4.2020.1482-1489>
- [17] Kesharwani, P., Chadar, R., Sheikh, A., & Rizg, WY., Safhi, AY., 2022. CD44-Targeted Nanocarrier for Cancer Therapy. *Front. Pharmacol.* 12, 800481. <https://doi.org/10.3389/fphar.2021.800481>
- [18] Naike Ye, Zekai Yang, Yuchen Liu., 2022 Applications of density functional theory in COVID-19 drug modeling, *Drug Discovery Today*. 27 (5) 1411-1419. <https://doi.org/10.1016/j.drudis.2021.12.017>.
- [19] Subramanian, N., Sundaraganesan, N., Sudha, S., Aroulmoji, V., Sockalingam, G.D., Bergamin, M., 2011. Experimental and theoretical investigation of the molecular and electronic structure of anticancer drug camptothecin, *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 78 (3) 1058-1067. <https://doi.org/10.1016/j.saa.2010.12.049>.
- [20] Iqar Ahmad., Vilas Jagatap., & Harun Patel. 2023. Chapter 17 - Application of density functional theory (DFT) and response surface methodology (RSM) in drug discovery, *Drug Discovery update*, 371-392. <https://doi.org/10.1016/B978-0-323-90593-0.00004-6>.
- [21] Zhang, Y., Tan, C.Y., Spjut, R.W., Fuchs, J.R., Kinghorn, A.D., Rakotondraibe, L.H. 2020. Specialized metabolites of the United States lichen *Niebla homalea* and their anti proliferative activities. *Phytochemistry*, 180, 112521. <https://doi.org/10.1016/j.phytochem.2020.112521>
- [22] Lai, D., Odimegwu, D.C., Esimone, C.; Grunwald, T.; Proksch, P. Phenolic compounds with in vitro activity against

- respiratory syncytial virus from the Nigerianlichen *Ramalina farinacea*. *Planta Med.* **2013**, 79, 1440–1446.
- [23] Angelini, MM., Akhlaghpour, M., Neuman, BW., Buchmeier, MJ. 2013. Severe acute respiratory syndrome coronavirus nonstructural proteins 3, 4, and 6 induce double-membrane vesicles. *mBio.* 4(4) e00524-13. <https://doi.org/10.1128/mbio.00524-13>
- [24] Berman, H. M., Battistuz, T., Bhat, T. N., Bluhm, W. F., Bourne, P. E., Burkhardt, K., Feng, Z., Gilliland, G. L., Iype, L., Jain, S., Fagan, P., Marvin, J., Padilla, D., Ravichandran, V., Schneider, B., Thanki, N., Weissig, H., Westbrook, J. D., Zardecki, C. 2002. The protein data bank. *Acta Crystallographica. Section D, Biological Crystall.*
- [25] Daina, A., Michielin, O., Zoete, V., (2017). Swiss ADME: A free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules. *Scientific Reports*, 7, 42717–42713. <https://doi.org/10.1038/srep42717>.
- [26] Frisch, A., Chandra, P., Smiley, D, Peng, L., Rizzo, M., Gatliffe, C., Hudson, M., Mendoza, J., Johnson, R., Lin, E., Umpierrez G E., 2010. Prevalence and clinical outcome of hyperglycemia in the perioperative period in non cardiac surgery. *Diabetes Care.* Aug;33(8) 1783-8. <https://doi.org/10.2337/dc10-0304>
- [27] Becke, A. D., 1993. A new mixing of Hartree-Fock and local density-functional theories. *The Journal of Chemical Physics*, 98(2)1372–1377. <https://doi.org/10.1063/1.464304>
- [28] Lee, C., Yang, W., Parr, R. G., 1988. Development of the Colle-Salvetti correlation-energy formula into a functional of the electron density. *Physical Review B*, 37(2) 785–789. <https://doi.org/10.1103/PhysRevB.37.785>
- [29] Trott, O., Olson, A. J., 2010. Auto Dock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of computational chemistry*, 31(2) 455-461.
- [30] Bowers, K. J., Chow, E., Xu, H., Dror, R. O., Eastwood, M. P., Gregersen, B. A., Klepeis, J. L., Kolossvary, I., Moraes, M. A., Sacerdoti, F. D., Salmon, J. K., Shan, Y., Shaw, D. E. 2006. Scalable algorithms for molecular dynamics simulations on commodity clusters [Paper presentation]. *Proceedings of the 2006 ACM/IEEE Conference on Supercomputing, SC'06, November.* <https://doi.org/10.1145/1188455.1188544>.
- [31] Shivakumar, D., Williams, J., Wu, Y., Damm, W., Shelley, J., Sherman, W. 2010. Prediction of absolute solvation free energies using molecular dynamics free energy perturbation and the OPLS force field. *Journal of Chemical Theory and Computation*, 6(5)1509–1519. <https://doi.org/10.1021/ct900587b>.
- [32] Thirumalaisamy, R, Aroulmoji, V., Iqbal, M.N., Deepa, M., Sivasankar, C., Khan, R., Selvankumar, T. 2021. Molecular Insights of the Hyaluronic Acid-Hydroxychloroquine Conjugate as a Promising Drug in Targeting SARS-CoV-2 Viral Proteins, *Journal of Molecular Structure* 1238, Aug 15, 130457. <https://doi.org/10.1016/j.molstruc.2021.130457>
- [33] Noor, H, A., Ikram, T., Rathinavel, S., Kumarasamy, M., NasirIqbal, Bashir, Z., 2021. Immunomodulatory and anti-Cytokine Therapeutic Potential of Curcumin and Its Derivatives for Treating COVID-19–A Computational Modeling, *Journal of Biomolecular Structure and Dynamics* 40(13) 5769-5784. <https://doi.org/10.1080/07391102.2021.1873190>
- [34] Samuel Genheden, and Ulf Ryde, 2015. Ulf the MM/PBSA and MM/GBSA Methods to Estimate Ligand-Binding Affinities, *Expert Opinion on Drug Discovery* 10(5) 449–61. 67.
- [35] Wang, Yu-Jen Jou, Su-Hua Huang, Li-Hsin Hsiao, Lei Wan, Ying-Ju Lin, Szu-Hao Kung, Cheng-Wen Lin, 2016. SARS Coronavirus Papain-like Protease Inhibits the TLR7 Signaling Pathway through Removing Lys63-Linked Polyubiquitination of TRAF3 and TRAF6, *International Journal of Molecular Sciences* 17(5) 678.

- [36] <https://doi.org/10.3390/ijms17050678>
Malone, B., & Campbell, EA, (2021). Molnupiravir: coding for catastrophe. *Nat Struct Mol Biol* 28(9) 706–711. <https://doi.org/10.1038/s41594-021-00657-8>
- [37] Baez-Santos, Y. M., St John, S. E., Mesecar, A. D. (2015). The SARS coronavirus papain-like protease: structure, function and inhibition by designed antiviral compounds. *Antivir. Res.* 115 21–38. <https://doi.org/10.1016/j.antiviral.2014.12.015>
- [38] Emmie de Wit., Feldmann, F., Cronin, J., Jordan, R., Okumura, A., Thomas, T., Scott, D., Cihlar, T., & Feldmann, H., 2020. Prophylactic and therapeutic remdesivir (GS625 5734) treatment in the Rhesus macaque model of MERS-CoV infection. *Proc Natl Acad Sci U S A.* 117, 6771–6776. <https://doi.org/10.1073/pnas.1922083117>.
- [39] Sheahan, T.P., Sims, A.C., Graham, R.L., Menachery, V.D., Gralinski, L.E., Case, J.B., Leist, S.R., Pyrc, K., Feng, J.Y., Trantcheva, I. 2017. Broad-spectrum Journal Pre-proof 33 antiviral GS-5734 inhibits both epidemic and zoonotic coronaviruses. *Sci. Transl. Med.* 9 <https://doi.org/10.1126/scitranslmed.aal3653>
- [40] Choy, K.T., Wong, A.Y., Kaewpreedee, P., Sia, S.F., Chen, D., Hui, K.P.Y., Chu, D.K.W., Chan, M.C.W., Cheung, P.P., Huang, X., 2020. Remdesivir, lopinavir, emetine, and homoharringtonine inhibit SARS-CoV-2 replication in vitro. *Antivir. Res.* 178, 104786. <https://doi.org/10.1016/j.antiviral.2020.104786>
- [41] Kotaki, T., Xie, X., Shi, P.Y., & Kameoka, M. A., (2021). PCR amplicon-based SARS CoV-2 replicon for antiviral evaluation. *Sci. Rep.* 11, 2229. <https://doi.org/10.1038/s41598-021-82055-0>
- [42] Mohan Mani., Vellusamy, M., Rathinavel, T., Vadivel, P., Dauchez, M., Khan, R., Aroulmoji, V., 2024 a. In silico validation of hyaluronic acid - drug conjugates based targeted drug delivery for the treatment of COVID-19. *J Biomol Struct Dyn.* Mar 27, 1–15.
- [43] Mohan.M., Rekha, P., Gokulraj, P., Azhagar Samy, P., Thirumalaisamy, R. 2024 b. Molecular Studies of Antiviral Drug Atazanavir and Hyaluronic Acid-Atazanavir conjugate as Novel Drugs to Target SARS-CoV-2 Viral Proteins. *Int. J. Adv. Sci. Eng.* 10 (3) 3581–3592. <https://doi.org/10.29294/IJASE.10.3.2024.3581-3592>
- [44] Lapčík, Jr. L., Lapčík, L., De Smedt, S., Demeester, J., Chabreck, P., 1998. Hyaluronan: preparation, structure, properties and applications. *Chemical Reviews*, 98, 2663–2684.
- [45] Schanté, CE., Zuber, G., Herlin, C., Vandamme, TF, 2011. Chemical modifications of hyaluronic acid for the synthesis of derivatives for a broad range of biomedical applications. *Carbohydrate Polymers*, 85, 469–489 <https://doi.org/10.1016/j.carbpol.2011.03.019>
- [46] Goodarzi, N., Varshochian, R., Kamalinia, G., Atyabi, F., Dinarvand, R., 2013. A review of polysaccharide cytotoxic drug conjugates for cancer therapy, *Carbohydrate Polymers*, 92, 1280–1293. <https://doi.org/10.1016/j.carbpol.2012.10.036>
- [47] Muthu Raj, S., Manohar, M., Mohan, M. 2021. Potential binding efficiency of antiviral drug Lopinavir targeted to the catalytic dyad, His41 - Cys145 of SARS CoV-2 Main protease. *International Journal of Life Sciences and Research* 3 (1) 22–25.
- [48] Ki Young Choi, Kyung Hyun Min, Hong Yeol Yoon, Kwangmeyung Kim, Jae Hyung Park, Ick Chan Kwon, Kuiwon Choi, Seo Young Jeong., 2011. PEGylation of hyaluronic acid nanoparticles improves tumor targetability in vivo. *Biomaterials* 32:1880–9. <https://doi.org/10.1016/j.biomaterials.2010.11.010>
- [49] Guan, L., Yang, H., Cai, Y., Sun, L., Di, P., Li, W., Liu, G., Tang, Y., 2018. ADMET-score - a comprehensive scoring function for evaluation of chemical drug-likeness. *Med chem. comm.* 30, 10(1)148–157. <https://doi.org/10.1039/C8MD00472B>
- [50] Ahmet Aktaş., Burak Tüzün., Rukiye Aslan., Koray Sayın., Hilmi Ataseven. 2020. New anti-viral drugs for the

- treatment of COVID-19 instead of Favipiravir, Journal of Biomolecular Structure and Dynamics, 39(18) 7263–7273.
<https://doi.org/10.1080/07391102.2020.1806112>
- [51] Williams, J. A., Hyland, R., Jones, B. C., Smith, D. A., Hurst, S., Goosen, T. C., Peterkin, V., Koup, J. R., Bal, S. E., 2004. Drug-Drug Interactions For Udp-Glucuronosyl transferase Substrates: A Pharmacokinetic Explanation For Typically Observed Low Exposure (AUC_i/AUC) Ratios, Drug Metab. Dispos. 32:1201–1208.
<https://doi.org/10.1124/dmd.104.000794>
- [52] Thirumalaisamy, R., Aroulmoji, V., Iqbal, M. N., Saride, S., Bhuvaneswari, M., Deepa, M. Siva sankar, C., Khan, R., 2023 Molecular Insights of Hyaluronic Acid - Ethambutol and Hyaluronic Acid - Isoniazid Drug Conjugates act as Promising Novel Drugs for the Treatment of Tuberculosis, Journal of Biomolecular Structure & Dynamics 41(8)3562-3573.
<https://doi.org/10.1080/07391102.2022.2051748>
- [53] Vasuki Selvaraj., Thirumalaisamy Rathinave.I., Subramanian Ammash., Muhammad Nasir Iqbal., 2022. Polyphenolic Phytochemicals Exhibit Promising SARS-COV-2 Papain Like Protease (PLpro) Inhibition Validated through a Computational Approach, Polycyclic Aromatic Compounds, 43(6)5545–5566.
<https://doi.org/10.1080/10406638.2022.2103578>
- [54] Mohan, M., Aroulmoji, V., Manohar, M., Ganesh, P., Rekha, P., Shreyas Saride., Riaz Khan., 2022. SARS-CoV-2: Pathophysiology, Prophylaxis and Treatment Opportunities -A Current Review. Int. J. Adv. Sci. Eng.9 (1) 2637-2657
<https://doi.org/10.29294/IJASE.9.1.2022.2637-2657>
- [55] Mishra, S.S., Kumar, N., Singh, H.P., Ranjan, S., Sharma, C.S., 2018. In silico pharmacokinetic, bioactivity and toxicity study of some selected anti-asthmatic agents. Int. J. Pharm. Sci. Drug Res. 10, 278–282.
- [56] Ramasamy Srinivasan., Desingu Kamalanathan., Thirumalaisamy Rathinavel., Muhammad Nasir Iqbal., Gnanendra Shanmugam. 2023. Anti-cancer potentials of aervine validated through in silico molecular docking, dynamics simulations, pharmacokinetic prediction and in vitro assessment of caspase – 3 in SW480 cell line, Molecular Simulation, 49(8) 799–815.
<https://doi.org/10.1080/08927022.2023.2193646>
- [57] Campillo, N., Falcones, B., Otero, J., Colina, R., Gozal, D., Navajas, D., Farré, R., Almendros I., 2019. Differential Oxygenation in Tumor Microenvironment Modulates Macrophage and Cancer Cell Crosstalk: Novel Experimental Setting and Proof of Concept. Front. Oncol. 9 43.
<https://doi.org/10.3389/fonc.2019.00043>
- [58] Ian Fleming.,1976. Frontier orbitals and organic chemical reactions.Wiley.
- [58] Deepak Vijay, K., Prakasam, A., Arivazhagan, R., Anbarasan, P. M. 2023.Insights into the structural, electronic, quantum chemical properties and molecular docking studies on novel NAMPT inhibitor molecule. Chemical Physics Impact, 7,100395.
<https://doi.org/10.1016/j.chphi.2023.100395>
- [59] Wolinski, K., Hinton, J. F., Pulay, P.1990. Efficient implementation of the gauge-independent atomic orbital method for NMR chemical shift calculations. Journal of the American Chemical Society, 112(23)8251–8260.
<https://doi.org/10.1021/ja00179a005>
- [60] Lewis, D. F. V., Ioannides, C., Parke, D. V. 1994. Interaction of a series of nitriles with the alcohol-inducible isoform of P450: Computer analysis of structure—activity relationships. Xenobiotica, 24(5) (1994) 401–408.
<https://doi.org/10.3109/00498259409043243>
- [61] Aihara, J. 1999. Reduced HOMO–LUMO Gap as an Index of Kinetic Stability for Polycyclic Aromatic Hydrocarbons.The Journal of Physical Chemistry A, 103(37)7487–7495.
<https://doi.org/10.1021/jp990092i>
- [62] Deepak vijay, K., Prakasam, A. 2024. Exploring the effects of mono-

- bromination on hole-electron transport and distribution in dibenzofuran and dibenzothiophene isomers: a first-principles study. *Journal of Molecular Modeling*, 30(6) 171.
<https://doi.org/10.1007/s00894-024-05966-5>
- [63] Politzer, P., Abu-Awwad, F., 1998. A comparative analysis of Hartree-Fock and Kohn-Sham orbital energies. *Theoretical Chemistry Accounts: Theory, Computation, and Modeling (Theoretica Chimica Acta)*, 99(2) 83–87.
<https://doi.org/10.1007/s002140050307>
- [64] Mulliken, R. S. A., 1934. New Electroaffinity Scale; Together with Data on Valence States and on Valence Ionization Potentials and Electron Affinities. *The Journal of Chemical Physics*, 2(11) 782–793.
<https://doi.org/10.1063/1.1749394>
- [65] Padmanabhan, J., Parthasarathi, R., Subramanian, V., Chattaraj, P. K., 2007. Electrophilicity-Based Charge Transfer Descriptor. *The Journal of Physical Chemistry A*, 111(7)1358–1361.
<https://doi.org/10.1021/jp0649549>
- [66] Chattaraj, P. K., Sarkar, U., Roy, D. R. 2006. Electrophilicity Index. *Chemical Reviews*, 106(6) 2065–2091.
<https://doi.org/10.1021/cr040109f>
- [67] Ghosh, D. C., Islam, N. 2011. Whether electronegativity and hardness are manifest two different descriptors of the one and the same fundamental property of atoms-A quest. *International Journal of Quantum Chemistry*, 111(1)40–51.
<https://doi.org/10.1002/qua.22415>
- [68] Peljo, P., Girault, H. H. 2018. Electrochemical potential window of battery electrolytes: the HOMO–LUMO misconception. *Energy & Environmental Science*, 11(9), 2306–2309.
<https://doi.org/10.1039/C8EE01286E>
- [69] Mekenyan, O. G., Ankley, G. T., Veith, G. D., Call, D. J., 1994. QSARs for photo induced toxicity: I. Acute lethality of polycyclic aromatic hydrocarbons to *Daphnia magna*. *Chemosphere*, 28(3) 567–582.
[https://doi.org/10.1016/0045-6535\(94\)90299-2](https://doi.org/10.1016/0045-6535(94)90299-2)
- [70] Schnurpfeil, G., Sobbi, A. K., Spiller, W., Kliesch, H., Wöhrle, D. 1997. Photo-oxidative Stability and its Correlation with Semi-empirical MO Calculations of Various Tetraazaporphyrin Derivatives in Solution. *Journal of Porphyrins and Phthalocyanines*, 01(02)159–167.
[https://doi.org/10.1002/\(SICI\)1099-1409\(199704\)1:2<159::AID-JPP19>3.0.CO;2-8](https://doi.org/10.1002/(SICI)1099-1409(199704)1:2<159::AID-JPP19>3.0.CO;2-8)
- [71] Muhammad I. Ismail., Hanan M. Ragab., Adnan A. Bekhit., Tamer M. Ibrahim., 2021. Targeting multiple conformations of SARS-CoV2 Papain-Like Protease for drug repositioning: An in-silico study, *Computers in Biology and Medicine*. 131,104295.
<https://doi.org/10.1016/j.compbiomed.2021.104295>
- [72] Hosseini, SA., Zahedipour, F., Mirzaei, H., Kazemi Oskuee, R. 2021. Potential SARS-CoV-2 vaccines: Concept, progress, and challenges. *Int Immuno Pharmacol*. Aug;97, 107622.
<https://doi.org/10.1016/j.intimp.2021.107622>
- [73] Ferreira, G.M., Pillaiyar, T., Hirata, M.H., Poso, A., Kronenberger, T., 2022. Inhibitor induced conformational changes in SARS-COV-2 papain-like protease. *Sci Rep* 12, 11585.
<https://doi.org/10.1038/s41598-022-15181-y>